THE UNIVERSITY OF SYDNEY
MATH2921 Tutorial Sheet 9 – Solution (Week 10)

Semester 1, 2025 | Vector Calculus and Differential Equations (Advanced)

Suggestion: try all problems, ideally before tutorial class. Feel free to use e-visual supports in the process, although it is ultimately important knowing how to proceed with just pen and paper.

1. Find a particular solution to the following equations:

   (a) $y'' - 3y' + 2y = e^{3x}(-1 + 2x + x^2)$;

   (b) $y'' - 8y' + 16y = 2e^{4x}$;

   (c) $6y'' + y' - y = e^{-x/2}(29 - 20x)$.

Solution:

(a) The characteristic equation for the complementary equation is

$$r^2 - 3r + 2 = (r-2)(r-1) = 0.$$

So the general solution to the complementary equation is

$$y_h(x) = c_1 e^{2x} + c_2 e^x.$$

None of these terms appear on the right hand side, so we guess that a particuar solution is of the form

$$y_p = e^{3x}(a_0 + a_1 x + a_2 x^2).$$

Differentiating we have

$$\begin{aligned} y_p &= e^{3x}(a_0 + a_1 x + a_2 x^2) \\ y_p' &= e^{3x}(3a_0 + a_1 + (3a_1 + 2a_2)x + 3a_2 x^2) \\ y_p'' &= e^{3x}(9a_0 + 6a_1 + 2a_2 + (9a_1 + 12a_2)x + 9a_2 x^2) \end{aligned}$$

Plugging this into the ODE gives

$$(2a_0 + 3a_1 + 2a_2 + (2a_1 + 6a_2)x + 2a_2)e^{3x} = e^{3x}(-1 + 2x + x^2),$$

so, equating coefficients, we need to solve the system

$$\begin{pmatrix} 2 & 3 & 2 \\ 0 & 2 & 6 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$$

which has solution

$$a_2 = \frac{1}{2} \quad a_1 = -\frac{1}{2} \quad a_0 = -\frac{1}{4}$$

so we have that a particular solution to the ODE is

$$y_p(x) = \frac{1}{4} e^{3x} \left(2x^2 - 2x - 1\right).$$

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(b) The characteristic equation for the complementary equation is

$$r^2 - 8r + 16 = (r-4)^2,$$

and so the general solution to the complementary equation is

$$y_h(x) = e^{4x}(c_1 + c_2 x).$$

Since e^{4x} is on the right hand side, we need to choose as our guess $y_p(x) = x^s e^{4x}$ where s is the lowest power so that $x^s e^{4x}$ isn't a solution to the complementary equation. In this case $s = 2$. So our guess is

$$y_p(x) = Ax^2 e^{4x}$$

Differentiating we have

$$\begin{aligned} y_p &= Ax^2 e^{4x} \\ y_p' &= A(2x + 4x^2)e^{4x} \\ y_p'' &= A(2 + 16x + 16x^2)e^{4x} \end{aligned}$$

Plugging this into the ODE and reducing gives

$$2Ae^{4x} = 2e^{4x},$$

and so we see that $A = 1$, and our particular solution is

$$y_p(x) = x^2 e^{4x}.$$

(c) The characteristic equation for the complementary equation is

$$6r^2 + r - 1 = (2r+1)(3r-1) = 0$$

So the roots are $r = -\frac{1}{2}$ and $\frac{1}{3}$ and the general solution to the homogeneous equation is

$$y_h(x) = c_1 e^{-x/2} + c_2 e^{x/3}$$

Our guess will be of the form

$$x^s e^{-x/2}(a_0 + a_1 x)$$

where s is chosen to make sure that no term is a solution to the homogeneous equation. In this case we need $s = 1$ so our guess will be

$$y_p(x) = xe^{-x/2}(a_0 + a_1 x).$$

Differentiating gives

$$\begin{aligned} y_p(x) &= e^{-x/2}(a_0 x + a_1 x^2) \\ y_p'(x) &= e^{-x/2}\left(a_0 + \left(2a_1 - \frac{a_0}{2}\right)x - \frac{a_1 x^2}{2}\right) \\ y_p''(x) &= e^{-x/2}\left(2a_1 - a_0 + \left(\frac{a_0}{4} - 2a_1\right)x + \frac{a_1 x^2}{4}\right). \end{aligned}$$

Plugging into the differential equation we get

$$(-5a_0 + 12a_1 - 10a_1 x)e^{-x/2} = e^{-x/2}(29 - 20x),$$

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so cancelling the exponential term and equating coefficients of like powers of x we need to solve the system

$$\begin{pmatrix} -5 & 12 \\ 0 & -10 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} 29 \\ -20 \end{pmatrix}$$

which has solution

$$a_1 = 2 \quad a_0 = -1.$$

So our particular solution to the ODE is

$$y_p(x) = e^{-x/2} \left(2x^2 - x\right).$$

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2. Prove Abel's formula for a general 3×3 linear first order system. That is to show the Wronskian of a set of three solutions to

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}' = A(x) \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

where $A(x)$ is C^1 on some interval (a, b), satisfies

$$W(x) = W(x_0) e^{\int_{x_0}^{x} \operatorname{tr}(A(t))\, dt}$$

for $x \in (a, b)$.

Solution:

We let $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}$ be a set of solutions to $\mathbf{y}' = A(x)\mathbf{y}$, and let

$$A(x) := \begin{pmatrix} a(x) & b(x) & c(x) \\ d(x) & e(x) & f(x) \\ g(x) & h(x) & j(x) \end{pmatrix}.$$

Then the Wronskian is defined as

$$W(\mathbf{u}, \mathbf{v}, \mathbf{w})(x) = \begin{vmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{vmatrix} =$$
$$u_1 v_2 w_3 + v_1 w_2 u_3 + w_1 u_2 v_3 - w_1 v_2 u_3 - v_1 u_2 w_3 - u_1 w_2 v_3$$

Differentiating gives

$$\begin{aligned}
W'(x) &= u_1' v_2 w_3 + u_1 v_2' w_3 + u_1 v_2 w_3' + v_1' w_2 u_3 + v_1 w_2' u_3 + v_1 w_2 u_3' \\
&\quad + w_1' u_2 v_3 + w_1 u_2' v_3 + w_1 u_2 v_3' - w_1' v_2 u_3 - w_1 v_2' u_3 - w_1 v_2 u_3' \\
&\quad - v_1' u_2 w_3 - v_1 u_2' w_3 - v_1 u_2 w_3' - u_1' w_2 v_3 - u_1 w_2' v_3 - u_1 w_2 v_3' \\
&= (a u_1 + b u_2 + c u_3) v_2 w_3 + u_1 (d v_1 + e v_2 + f v_3) w_3 + u_1 v_2 (g w_1 + h w_2 + j w_3) \\
&\quad + (a v_1 + b v_2 + c v_3) w_2 u_3 + v_1 (d w_1 + e w_2 + f w_3) u_3 + v_1 w_2 (g u_1 + h u_2 + j u_3) \\
&\quad + (a w_1 + b w_2 + c w_3) u_2 v_3 + w_1 (d u_1 + e u_2 + f u_3) v_3 + w_1 u_2 (g v_1 + h v_2 + j v_3) \\
&\quad - (a w_1 + b w_2 + c w_3) v_2 u_3 - w_1 (d v_1 + e v_2 + f v_3) u_3 - w_1 v_2 (g u_1 + h u_2 + j u_3) \\
&\quad - (a v_1 + b v_2 + c v_3) u_2 w_3 - v_1 (d u_1 + e u_2 + f u_3) w_3 - v_1 u_2 (g w_1 + h w_2 + j w_3) \\
&\quad - (a u_1 + b u_2 + c u_3) w_2 v_3 - u_1 (d w_1 + e w_2 + f w_3) v_3 - u_1 w_2 (g v_1 + h v_2 + j v_3) \\
&= (a + e + j)(u_1 v_2 w_3 + v_1 w_2 u_3 + w_1 u_2 v_3 - w_1 v_2 u_3 - v_1 u_2 w_3 - u_1 w_2 v_3) \\
&= \operatorname{tr}(A(x))\, W(x)
\end{aligned}$$

Solving the initial value problem for some $x_0 \in (a, b)$ gives the result

$$W(x) = W(x_0) e^{\int_{x_0}^{x} \operatorname{tr}(A(t))\, dt}.$$

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3. Suppose that $\alpha \neq 0$ and k is a positive integer. Most calculus books compute the integrals $\int x^k e^{\alpha x}$ by integrating by parts k times. This exercise shows another method. Let

$$y = \int e^{\alpha x} P(x) dx$$

with

$$P(x) = p_0 + p_1 x + p_2 x^2 + \cdots + p_k x^k, \quad (\text{where } p_k \neq 0)$$

(a) Show that $y = e^{\alpha x} u$ where

$$u' + \alpha u = P(x).$$

(b) Show that the equation in part (a) has a particular solution of the form

$$u_p = A_0 + A_1 x + \cdots + A_k x^k,$$

where $A_k, A_{k-1}, \ldots, A_0$ can be computed successively in terms of $p_k, p_{k-1}, \ldots p_0$.

(c) Conclude that

$$\int e^{\alpha x} P(x) dx = (A_0 + A_1 x + \cdots + A_k x^k) e^{\alpha x} + C$$

where C is a constant of integration.

(d) Use this to evaluate

$$\int e^x (x+1)^2 dx.$$

Solution:

(a) Suppose that $y = e^{\alpha x} u$, we have

$$e^{\alpha x} u = \int e^{\alpha x} P(x) dx.$$

Differentiating both sides gives

$$\alpha e^{\alpha x} u + e^{\alpha x} u' = e^{\alpha x} P(x)$$

and cancelling the exponentials gives the result:

$$\alpha u + u' = P(x).$$

(b) We write

$$u_p = A_k x^k + A_{k-1} x^{k-1} + \cdots + A_0.$$

Assuming u_p solves the equation from (a), we have that

$$u_p' = k A_k x^{k-1} + (k-1) A_{k-1} x^{k-2} + \cdots + A_1$$

so

$$u_p' + \alpha u_p = \alpha A_k x^k + (\alpha A_{k-1} + k A_k) x^{k-1} + \cdots + (\alpha A_0 + A_1)$$

Using the fact that this $= P(x)$ and equating coefficients of like powers of x enables us to recursively solve for the A_ks in terms of α and the p_k's. We have

$$A_k = \frac{p_k}{\alpha}, \quad A_{k-1} = \frac{p_{k-1} - k A_k}{\alpha} = \frac{p_{k-1}}{\alpha} - \frac{k p_k}{\alpha^2}, \quad \text{etc...}$$

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(c) We have that the general solution to $u' + \alpha u = P(x)$ is given by

$$u = Ce^{-\alpha x} + u_p$$

and we have that $\int e^{\alpha x} P(x) dx = y = e^{\alpha x} u \Rightarrow y = C + e^{\alpha x} u_p$

(d) Here we have that $\alpha = 1$ and we suppose that we have a particular solution of the form

$$\begin{aligned} u_p &= A_2 x^2 + A_1 x + A_0 \Rightarrow \\ u_p' &= 2A_2 x + A_1 \Rightarrow \\ u_p' + u_p &= A_2 x^2 + (A_1 + 2A_2)x + (A_1 + A_0) = x^2 + 2x + 1 \Rightarrow \\ A_2 &= 1 \quad A_1 = 0 \quad A_0 = 1. \end{aligned}$$

Putting this together with the first 3 parts to the question we have that

$$\int e^x (x+1)^2 dx = e^{\alpha x} u_p + C = e^x (x^2 + 1) + C.$$

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4. Find the general solution to the following equations:

(a) $y'' + 3y' + 2y = 7\cos x - \sin x$;

(b) $y'' + 4y = -12\cos 2x - 4\sin 2x$;

(c) $y'' + 2y' + y = 8x^2\cos x - 4x\sin x$.

Solution:

(a) The characteristic equation of the complementary equation is

$$(r+2)(r+1) = 0,$$

so the solution to the complementary equation is

$$y_h(x) = c_1 e^{-2x} + c_2 e^{-x}.$$

To find a particular solution, we guess that it is of the form

$$y_p(x) = A\cos x + B\sin x.$$

Substituting this into the ODE and collecting the coefficients of $\cos x$ and $\sin x$ we get the system

$$\begin{pmatrix} 1 & 3 \\ -3 & 1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \end{pmatrix}$$

Which has solution $a = 1$ and $b = 2$. So the general solution to the ODE is

$$y(x) = C_1 e^{-2x} + C_2 e^{x} + \cos x + 2\sin x.$$

(b) The (real) general solution to the complementary equation is

$$y_h(x) = c_1\cos 2x + c_2\sin 2x$$

which both appear on the right hand side. So we need to choose as our guess for the particular equation

$$y_p(x) = x(A\cos 2x + B\sin 2x).$$

Substituting this into the ODE and collecting the coefficients of $\sin 2x$ and $\cos 2x$. we get the (particularly easy to solve) system

$$\begin{pmatrix} 0 & 4 \\ -4 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -12 \\ -4 \end{pmatrix},$$

whence we see that $b = -3$ and $a = 1$. So the general solution to the ODE is

$$y(x) = C_1\cos 2x + C_2\sin 2x + x(\cos 2x - 3\sin 2x).$$

(c) The general solution to the complementary equation is

$$y_h(x) = e^{-x}(c_1 + c_2 x).$$

We make the guess for the particular solution

$$y_p(x) = \left(a_0 + a_1 x + a_2 x^2\right)\cos x + \left(b_0 + b_1 x + b_2 x^2\right)\sin x.$$

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Substituting into the ODE and collecting the coefficients of $x^k \sin 2x$ and $x^k \cos 2x$ for $k = 0, 1, 2$ we get the system

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 0 \\ 0 & 0 & 4 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 & 0 & 2 \\ -2 & -2 & 0 & 0 & 2 & 2 \\ 0 & -2 & -4 & 0 & 0 & 4 \\ 0 & 0 & -2 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ b_0 \\ b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 8 \\ 0 \\ -4 \\ 0 \end{pmatrix}$$

which has the solution

$$\begin{pmatrix} a_0 & a_1 & a_2 & b_0 & b_1 & b_2 \end{pmatrix} = \begin{pmatrix} -14 & 10 & 0 & -2 & -8 & 4 \end{pmatrix}$$

and so the general solution is

$$y(x) = e^{-x}(C_1 + xC_2) + (10x - 14)\cos x + (4x^2 - 8x - 2)\sin x.$$

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5. Consider the linear system of equations $\mathbf{y}' = A(x)\mathbf{y}$ where $\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ and $A(x)$ is the 2×2 matrix given below:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ -\frac{1}{x} & \frac{1}{x} + 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \quad x > 0 \tag{1}$$

(a) Compute $B(x) = \int A(x)dx$, you can choose the constants of integration to be 0 for each entry.

(b) Compute the eigenvalues and eigenvectors of $B(x)$ (in terms of x). Use this to compute the matrix exponential $e^{B(x)}$.

(c) Compute $B'(x)$, compute $\dfrac{d}{dx}\left(e^{B(x)}\right)$ and verify that it is not a fundamental solution matrix for eq. (1).

(d) It is a fact that the variation of parameters formula will hold when $\Phi(x)$ is any fundamental matrix solution, not just a principal one. Use this fact, plus the fact that a fundamental matrix solution to eq. (1) is

$$\Phi(x) = \begin{pmatrix} x+1 & e^x \\ 1 & e^x \end{pmatrix}$$

to find a particular solution to

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' - \begin{pmatrix} 0 & 1 \\ -\frac{1}{x} & \frac{1}{x} + 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x \\ x^2 \end{pmatrix}, \quad x > 0.$$

Solution:

(a)

$$\int A(x) = \begin{pmatrix} 0 & x \\ -\log x & x + \log x \end{pmatrix}.$$

(b) The eigenvalues of $B(x)$ are x and $\log x$. The corresponding eigenvectors are $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} x \\ \log x \end{pmatrix}$ respectively. Setting

$$P = \begin{pmatrix} 1 & x \\ 1 & \log x \end{pmatrix} \qquad P^{-1} = \frac{1}{\log x - x} \begin{pmatrix} \log x & -x \\ -1 & 1 \end{pmatrix}$$

we have that

$$B(x) = P \begin{pmatrix} x & 0 \\ 0 & \log x \end{pmatrix} P^{-1},$$

whence

$$e^{B(x)} = P \begin{pmatrix} e^x & 0 \\ 0 & x \end{pmatrix} P^{-1} = \frac{1}{\log x - x} \begin{pmatrix} e^x \log x - x^2 & x(x - e^x) \\ (e^x - x)\log x & x(\log x - e^x) \end{pmatrix}.$$

(c) We have that $B'(x) = A(x)$. Letting $D(x) = \begin{pmatrix} e^x & 0 \\ 0 & x \end{pmatrix}$ we have that $e^{B(x)} = PDP^{-1}$, so

$$\frac{d}{dx} e^{B(x)} = \frac{d}{dx}(PDP^{-1}) = P'DP^{-1} + PD'P^{-1} + PD(P^{-1})'.$$

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We have that

$$P' = \begin{pmatrix} 0 & 1 \\ 0 & \frac{1}{x} \end{pmatrix}, \quad D' = \begin{pmatrix} e^x & 0 \\ 0 & 1 \end{pmatrix}$$

and

$$(P^{-1})' = -P^{-1}P'P^{-1} = \frac{1}{(x - \log x)^2} \begin{pmatrix} \log x - 1 & 1 - \log x \\ \frac{1}{x} - 1 & 1 - \frac{1}{x}. \end{pmatrix}$$

You can check for yourself that $e^{B(x)}$ is not a fundamental solution matrix for eq. (1). For example the (1,1) entry of $\dfrac{d}{dx}e^{B(x)} - A(x)e^{B(x)}$ is

$$\frac{(\log x - 1)(e^x + x\log x - x - x^2)}{(x - \log x)^2} \neq 0.$$

(d) A particular solution will be given by

$$\Phi(x) \int^x \Phi^{-1}(t) \begin{pmatrix} t \\ t^2 \end{pmatrix} dt$$

We have that

$$\int^x \Phi^{-1}(t) \begin{pmatrix} t \\ t^2 \end{pmatrix} dt = \begin{pmatrix} x - \frac{x^2}{2} \\ -e^{-x}(x^2 + 3x + 2) \end{pmatrix}.$$

Thus a particular solution will be

$$y_p(x) = \begin{pmatrix} -\frac{1}{2}(1 + x)(x^2 + 4) \\ -2 - \frac{1}{2}x(4 + 3x) \end{pmatrix}.$$

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6. (Higher dimensional Euler equations) An equation of the form

$$x^n y^{(n)} + a_1 x^{n-1} y^{(n-1)} + \cdots + a_{n-1} x y' + a_n y = 0, \qquad x > 0 \tag{2}$$

where the a_i are constants is called an Euler or equidimensional equation.

(a) Show that if $x(t) = e^t$ and $Y(t) = y(x(t))$ then

$$\begin{aligned} x\frac{dy}{dx} &= \frac{dY}{dt}, \\ x^2\frac{d^2y}{dx^2} &= \frac{d^2Y}{dt^2} - \frac{dY}{dt}, \\ x^3\frac{d^3y}{dx^3} &= \frac{d^3Y}{dt^3} - 3\frac{d^2Y}{dt^2} + 2\frac{dY}{dt}. \end{aligned} \tag{3}$$

(b) Use the results in part (a) to write the general solution to

$$x^3 y''' + 5x^2 y'' + 2xy' - 2y = 0.$$

Solution:

(a) We first observe that $\frac{dx}{dt} = x$. Then by differentiating $Y(t) = y(x(t))$ with respect to t and applying the chain rule, we have

$$\frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{dy}{dx}x = \frac{dY}{dt}.$$

Differentiating both sides of the second equation again with respect to t we have

$$\begin{aligned} \frac{d^2y}{dx^2}x^2 + \frac{dy}{dx}x &= \frac{d^2Y}{dt^2} \Rightarrow \\ \frac{d^2y}{dx^2}x^2 &= \frac{d^2Y}{dt^2} - \frac{dY}{dt}. \end{aligned}$$

Finally, differentiating the last equation again with respect to t we have

$$\begin{aligned} \frac{d^3y}{dx^3}x^3 + 2\frac{d^2y}{dx^2}x^2 &= \frac{d^3Y}{dt^2} - \frac{d^2Y}{dt^2} \Rightarrow \\ \frac{d^3y}{dx^3}x^3 &= \frac{d^3Y}{dt^3} - 3\frac{d^2Y}{dt^2} + 2\frac{dY}{dt}. \end{aligned}$$

(b) Substituting the expressions in from Part (a), we have the the differential equation becomes

$$Y'''(t) + 2Y''(t) - Y'(t) - 2Y(t) = 0.$$

You can solve this in any number of ways that you like. The general solution is

$$Y(t) = c_1 e^t + c_2 e^{-t} + c_3 e^{-2t}$$

where c_i are arbitrary constants. Recalling that $Y(t) = y(e^t)$ we have that in terms of x the general solution becomes

$$y(x) = c_1 x + \frac{c_2}{x} + \frac{c_3}{x^2}.$$

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7. (Why it's called variation of parameters.) Traditionally, what is referred to as 'variation of parameters' in most ODEs courses (including the book by Trench) is not evidently recognisable as the method of integrating factors for systems of linear equations. This problem will derive the more traditional formula for second order linear ODEs (though one that is harder to remember). You are also encouraged to read the beginning of section 5.7 to find an even more general procedure. Suppose we were to consider the linear ODE

$$y'' + p(x)y' + q(x)y = f(x) \tag{4}$$

and suppose that $\{y_1, y_2\}$ formed a fundamental set of solutions to the complementary equation. We want to look for a solution of the form

$$y_p = u_1 y_1 + u_2 y_2$$

Here u_1 and u_2 are functions that depend on x to be determined so that y_p solves eq. (4) (hence the varying parameters).

(a) Suppose that u_1 and u_2 satisfy

$$u_1' y_1 + u_2' y_2 = 0. \tag{5}$$

Use the fact that y_p is a solution eq. (4) to arrive at a second (linear) condition that u_1' and u_2' must satisfy.

(b) Write these two linear equations as a system in the variables u_1' and u_2'. Then solve this system. From here you can integrate u_1' and u_2' to get what the varying parameters of y_1 and y_2 should be in y_p.

Solution:

(a) If we write $y_p = u_1 y_1 + u_2 y_2$ where y_1 and y_2 are linearly independent solutions to the complementary equation, then differentiating y_p twice we get

$$\begin{aligned}
y_p &= u_1 y_1 + u_2 y_2 \\
y_p' &= u_1' y_1 + u_1 y_1' + u_2' y_2 + u_2 y_2' = u_1 y_1' + u_2 y_2' \quad \text{(because of eq. (5))} \\
y_p'' &= u_1' y_1' + u_1 y_1'' + u_2' y_2' + u_2 y_2''
\end{aligned}$$

and since (by hypothesis) y_p is a solution to eq. (4), we have that

$$\begin{aligned}
f(x) &= u_1' y_1' + u_1 y_1'' + u_2' y_2' + u_2 y_2'' + p(x)\left(u_1 y_1' + u_2 y_2'\right) + q(x)\left(u_1 y_1 + u_2 y_2\right) \\
&= u_1\left(y_1'' + p(x) y_1' + q(x) y_1\right) + u_2\left(y_2'' + p(x) y_2' + q(x) y_2\right) + u_1' y_1 + u_2' y_2' \\
&= u_1' y_1' + u_2' y_2' \quad \text{(because } y_{1,2} \text{ solve the complementary equation).}
\end{aligned}$$

(b) We have the system

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ f \end{pmatrix}$$

Since $\{y_1, y_2\}$ form a fundamental set, their Wronskian is never zero, so you can invert the matrix above to solve the system. This gives the more classical formula for variation of constants (written in vector form)

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} \int \frac{-y_2 f}{y_1 y_2' - y_2 y_1'} \\ \int \frac{y_1 f}{y_1 y_2' - y_2 y_1'} \end{pmatrix}.$$

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As a final remark, you can observe that the denominator in the fractions above is the wronskian of the fundamental set $\{y_1 y_2\}$, which means that it is equal to $W(x_0)e^{-\int^x p(t)\,dt}$. Using this, you can simplify the expressions a little to

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \frac{1}{W(x_0)} \begin{pmatrix} -\int^x y_2(t) f(t) e^{\int^t p(s)\,ds}\,dt \\ \int^x y_1(t) f(t) e^{\int^t p(s)\,ds}\,dt \end{pmatrix}.$$

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8. Use variation of parameters (either way) to find general solution to the following

(a) $y'' - y = \frac{1}{\sinh x}, \quad x > 0;$
(b) $y'' - 3y' + 2y = \frac{1}{1-e^{-x}};$
(c) $y'' + 2y = e^x \sin x.$

Solution:

(a) For these, I will use the matrix exponential method, as I can't ever remember the other formula. We need to write the homogeneous problem out as a first order system, and then solve it. We have that the homogeneous ODE $y'' - y = 0$ is equivalent to (by setting $y = y_1$ and $y' = y_2$):

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} =: A \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}.$$

The general solution to this is then $e^{Ax} \begin{pmatrix} y_1(0) \\ y_2(0) \end{pmatrix}$. To compute the exponential, we need the eigenvalues and eigenvectors of A. The eigenvalues of A are ± 1 with eigenvectors $\begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$ respectively. Letting $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ be the matrix of eigenvectors, we have that $P^{-1} = \frac{1}{2}P$. Thus we have that

$$A = P \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} P^{-1}$$

and so

$$e^{Ax} = \begin{pmatrix} \frac{e^x + e^{-x}}{2} & \frac{e^x - e^{-x}}{2} \\ \frac{e^x - e^{-x}}{2} & \frac{e^x + e^{-x}}{2} \end{pmatrix} = \begin{pmatrix} \cosh x & \sinh x \\ \sinh x & \cosh x \end{pmatrix}.$$

Returning to the inhomogeneous equation, we write (a) as a system

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}' - A \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{1}{\sinh x} \end{pmatrix}$$

Multiplying both sides of this equation by $e^{A(-x)}$ we get

$$\left(e^{A(-x)} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right)' = \begin{pmatrix} -1 \\ \coth x \end{pmatrix}$$

Integrating both sides and solving for $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ we get

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{Ax} \begin{pmatrix} -x \\ \log(\sinh x) \end{pmatrix} + e^{Ax} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

where c_1 and c_2 are constants. The solution to the inhomogeneous equation will then be y_1, or the first row of the expression above. This is

$$y = c_1 \cosh x + c_2 \sinh c - x \cosh x + (\sinh x) \log(\sinh x).$$

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(b) The solution to the inhomogeneous equation is the first row of the following

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = e^{Ax} \begin{pmatrix} e^{-x} \\ 2e^{-x} + \log(1 - e^{-x}) \end{pmatrix} + e^{Ax} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \quad \text{where} \quad e^{Ax} = \begin{pmatrix} 2e^x - e^{2x} & e^{2x} - e^x \\ 2e^x - 2e^{2x} & 2e^{2x} - e^x \end{pmatrix}$$

and c_1 and c_2 are constants.

(c) The solution to the inhomogeneous equation is the first row of the following

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{4}e^x(\sin x - \cos x) \\ \frac{1}{2}e^x \sin x \end{pmatrix} + e^{Ax} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \quad \text{where} \quad e^{Ax} = \begin{pmatrix} \cos\sqrt{2}x & \frac{\sin\sqrt{2}x}{\sqrt{2}} \\ -\sqrt{2}\sin\sqrt{2}x & \cos\sqrt{2}x \end{pmatrix}$$

As an aside, you can solve this problem by making the ansatz that $y_p(x) = e^x(A\sin x + B\cos x)$ and then derive a system of equations for A and B. You will get the same answer.

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9. Find a particular solution to the following system of differential equations

$$\mathbf{y}' = \begin{pmatrix} 3 & 1 & -1 \\ 3 & 5 & 1 \\ -6 & 2 & 4 \end{pmatrix} \mathbf{y} + \begin{pmatrix} 3 \\ 6 \\ 3 \end{pmatrix}.$$

Solution:

We first find the fundamental solution matrix to the complementary equation

$$\mathbf{y}' = \begin{pmatrix} 3 & 1 & -1 \\ 3 & 5 & 1 \\ -6 & 2 & 4 \end{pmatrix} \mathbf{y} =: A\mathbf{y}.$$

We know that this is e^{At}. The eigenvalues of A are found to be $\{6, 6, 0\}$ with matrix of eigenvectors

$$P = \begin{pmatrix} -1 & 1 & 1 \\ 0 & 3 & -1 \\ 3 & 0 & 2 \end{pmatrix}$$

We thus have that

$$e^{At} = P \begin{pmatrix} e^{6t} & 0 & 0 \\ 0 & e^{6t} & 0 \\ 0 & 0 & 1 \end{pmatrix} P^{-1}.$$

Now with e^{At} given, we can proceed by an integrating factor. We want to solve

$$\mathbf{y}' - Ay = \mathbf{b}$$

where $\mathbf{b} = \begin{pmatrix} 3 \\ 6 \\ 3 \end{pmatrix}$. Multiplying on the left by e^{-At} reduces the equation to

$$\begin{aligned} \left(e^{-At}\mathbf{y}\right)' &= e^{-At}\mathbf{b} \Rightarrow \\ e^{-At}\mathbf{y} &= \int e^{-As}\mathbf{b}ds + \mathbf{c} \Rightarrow \\ \mathbf{y} &= e^{At}\int e^{-As}\mathbf{b}ds + e^{At}\mathbf{c} \end{aligned}$$

Since we just want a single particular solution, we choose $\mathbf{c} = 0$. When this is all computed out, and the dust settles, I got

$$\mathbf{y} = \begin{pmatrix} t - \frac{1}{3} \\ -t - \frac{7}{6} \\ 2t - \frac{1}{6} \end{pmatrix}.$$

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10. Find a particular solution to the following system of differential equations

$$\mathbf{y}' = \frac{1}{t^2-1}\begin{pmatrix} t & -1 \\ -1 & t \end{pmatrix}\mathbf{y} + t\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

given that

$$\Phi(t) = \begin{pmatrix} t & 1 \\ 1 & t \end{pmatrix}$$

is a fundamental matrix solution for the complementary equation:

$$\mathbf{y}' = \frac{1}{t^2-1}\begin{pmatrix} t & -1 \\ -1 & t \end{pmatrix}\mathbf{y}.$$

Solution:

We denote the following

$$A(t) := \frac{1}{t^2-1}\begin{pmatrix} t & -1 \\ -1 & t \end{pmatrix} \quad \text{and,} \quad \mathbf{b} = \begin{pmatrix} t \\ -t \end{pmatrix}.$$

To solve

$$\mathbf{y}' - A(t)\mathbf{y} = \mathbf{b}$$

we multiply on the left by $\Phi^{-1}(t)$, and then observe that the left hand side reduces, and can be integrated:

$$\begin{aligned} \left(\Phi^{-1}(t)\mathbf{y}\right)' &= \Phi^{-1}\mathbf{b} \Rightarrow \\ \Phi^{-1}(t)\mathbf{y} &= \int \Phi^{-1}(s)\mathbf{b}ds + \mathbf{c} \Rightarrow \\ \mathbf{y} &= \Phi(t)\int \Phi^{-1}(s)\mathbf{b}ds + \Phi(t)\mathbf{c} \end{aligned}$$

and since we just want a single particular solution, we can choose $\mathbf{c} = 0$. You can check for yourself that $\Phi^{-1}(t) = A(t)$, and that

$$\Phi(t)\int A(s)\mathbf{b}ds = (1-t)(t+\ln(t-1))\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

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